2.2.3 Gas exchange in mammals and plants

		Learning outcomes	Additional guidance
		Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	**(i)**	the relationships between cells, tissues and organs in the mammalian gas exchange system	To include the appearance and the histology of squamous epithelial cells in the alveoli, ciliated epithelial tissue, smooth muscle, cartilage and elastic fibres.
	(ii)	observations of tissues of the gas exchange system using microscopy	**PAG1** HSW4, HSW5, HSW6
(b)		the process of gas exchange in the alveoli	To include the roles of ventilation, epithelial tissue, smooth muscle, cartilage, elastic fibres, blood capillaries and surfactant in the establishment and maintenance of concentration gradients.
(c)		the parameters affecting pulmonary ventilation	To include consideration of tidal volume, breathing rate, vital capacity, residual volume, PEFR and FEV_1 (no detail of the use of a spirometer is required). **PAG10** *M0.1, M2.2, M2.3, M2.4*
(d)		how expired air resuscitation can be carried out on adults, children and babies in cases of respiratory arrest	To include reference to both manual and electronic methods. HSW3, HSW4, HSW5, HSW6
(e)		the process of gas exchange in terrestrial plants.	To include the diffusion of gases between the atmosphere and intercellular spaces of leaves via stomata and through the lenticels of stems.
(f)	**(i)**	the structure of stomata, their opening and closing	To include reference to changes in turgor and water potential of guard cells and the need for ATP.
	(ii)	the microscopic appearance of stomata	To include the appearance of stomata in the leaves of different terrestrial plants. **PAG1**